

The Algorithm of Being

Informational Ontology, Constraint, and the Self-Structuring Code

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Abstract

This work outlines a philosophical foundation for the informational age. Its central thesis is that physics, mathematics, computation, biology, cognition, consciousness, and artificial intelligence are not autonomous domains, but distinct projections of a single deeper architecture: **constrained relational dynamics**.

Information is interpreted not as data, but as topological restriction over the space of possibility. Structure emerges through distinction, constraint, selection, and stabilization. Meaning arises through semantic compression. Intelligence is adaptive modeling under constraint. Consciousness is not the center of ontology, but one of its highest expressions: a recursive semantic architecture that aligns internal models with the invariant structure of reality.

The aim is not to present a closed theory, but to formulate an ontological direction for a civilization increasingly shaped by information, computation, artificial intelligence, and the convergence of scientific and philosophical reflection.

1. Introduction: The Ontological Problem

Modern civilization commands extraordinary scientific, mathematical, and computational power. Yet it lacks a coherent ontology capable of integrating physics, information, cognition, consciousness, and artificial intelligence into a single conceptual field.

The age into which humanity is entering is not merely technological. It is ontological.

By *ontology* I mean not interpretation in the hermeneutic sense, but a framework describing:

- what fundamentally exists;
- how structure emerges from undifferentiated possibility;
- how order differs from noise;
- how meaning becomes possible;
- how intelligence arises;
- and how artificial systems fit into the architecture of being.

The guiding intuition of this work is:

Reality is not best understood as static substance, but as constrained, self-structuring informational process.

Physics increasingly speaks the language of information, entropy, symmetry, correlation, computation, and geometry emerging from deeper pre-geometric substrates — from Wheeler’s “it from bit” [1] to the thermodynamic derivation of the Einstein equations by Jacobson [2] and the holographic principle [3]. Mathematics describes invariant relations. Computer science formalizes representation, complexity, and

transformation. Biology reveals code, selection, regulation, and self-maintaining organization. Artificial intelligence externalizes cognitive processes before philosophy has fully clarified their nature.

These domains appear separate only at the surface. At a deeper level they converge toward one question:

How does structured meaning emerge from formal dynamics?

This text does not claim to answer the question completely. It attempts to give it a coherent direction, in the spirit of a framework where explanations are structural [11].

2. The Space of Algebras and the Informational Substrate

Statement

At the most fundamental level, reality should not be treated as ordinary space populated by objects, but as a **space of relations, distinctions, and possible transformations** — a relational field preceding metric geometry.

Definition

The **Space of Algebras** is a pre-metric reality consisting of relations and admissible transformations between them. It is not a set of points, but a category of possible distinctions and their compositions.

In this framework, metric structure, spacetime, and matter do not come first. They are stable regimes — attractors — of a deeper relational dynamics. Objects are not primitive. Objects are **stabilized patterns of relation**, emergent from the algebraic substrate through coarse-graining and constraint.

This construction is consonant with the causal set program [4], where spacetime emerges from locally finite partial orders, and with the relational stance in quantum gravity developed by Smolin [5]. The transition from discrete causal kinematics to continuum geometry, explored in the Causal Phase Geometry framework [6], demonstrates how Lorentzian metric, volume measure, and gauge connection arise jointly from a single microstructure: causal order plus $U(1)$ holonomies on oriented links.

Reflection

The intuitive image of a pre-given stage on which things appear is obsolete. At the most basic level there are distinguishable relations, possible transformations, constraints, and stable invariants.

The central question therefore shifts from:

What is the world made of?

to:

How does a stable world emerge from a relational field?

This does not require reducing physics to computation in the narrow, computationalist sense. It requires recognizing that physical reality, mathematical structure, and computation may all be expressions of a single formal-relational substrate [8, 9].

3. Information as Constraint

Statement

Information is not primarily data. Information is **restriction**.

Definition

Information is the measure of restriction imposed on the possible states of a system. Let Ω be the space of possible states. Information acts by reducing the admissible domain:

$$\Omega \rightarrow \Omega' \subset \Omega$$

Thus information is not merely something *contained* in a system. It is the principle by which the system becomes non-arbitrary.

Without informational restriction:

$$\text{Energy} \rightarrow \text{Undirected Entropy}$$

With informational restriction:

$$\text{Energy} \rightarrow \text{Stable Structure}$$

Reflection

Bits, signals, and symbols are surface manifestations of information. At a deeper level, information determines which states are admissible, which transformations persist, which correlations remain stable, and which trajectories survive.

If energy is the capacity for transformation, **information is the geometry of permitted transformation**.

This connects entropy, least action, computational efficiency, structural stability, and cognition under one general principle:

The world becomes intelligible because possibility is constrained.

4. Constraint as the Origin of Form

Statement

Structure emerges not from unrestricted possibility, but from **constrained generation**.

Definition

Self-structuring may be expressed as:

$$\text{Self-Structuring} = \text{Generation} + \text{Constraint} + \text{Selection}$$

Generation produces candidate forms. Constraint narrows the admissible space. Selection stabilizes coherent survivors.

Reflection

Unlimited possibility does not create order. It creates dissipation.

Without exclusion, pruning, boundary, and invariant preservation, no stable form can persist.

Constraint is therefore not the opposite of freedom. **Constraint is the condition under which meaningful freedom becomes possible.**

This principle appears across all ontological levels:

- Physical law constrains possible motion;
- Chemistry constrains possible bonding;
- Biology constrains viable forms;
- Language constrains meaning;
- Mathematics constrains proof;
- Cognition constrains attention;
- Artificial intelligence constrains hypothesis space.

Form is possibility made durable.

5. Motivation, Scaling, and the Logic of Development

Statement

The deepest question is not only how structure appears, but why structures develop, scale, and become capable of producing new levels of organization.

Definition

Systems do not tend toward maximal complexity in the naive sense. They tend toward **scaling**:

$$\text{Scaling} = \text{Expansion of Expressive Capacity} + \text{Preservation of Coherence}$$

Scaling is growth under pruning.

Reflection

A system that only accumulates collapses under its own weight. A system that only prunes becomes sterile.

Development requires the tension between expansion and preservation.

This pattern appears in physical optimization, biological evolution, cognitive compression, scientific law, language, and artificial intelligence.

A scientific theory does not merely contain more facts. It **compresses** more phenomena into fewer principles.

A mind does not process all distinctions. It **selects** the distinctions that matter.

An intelligent system develops by expanding what it can represent while preserving the coherence required to act.

6. Dynamic Balance and Structural Proportion

Statement

Development is not produced by order alone, nor by freedom alone. A system becomes capable of growth only when generation and constraint remain in a viable proportion.

Definition

Dynamic balance is the regulated proportion between generative freedom and constraining order that allows a system to increase expressive capacity without losing coherence:

$$\mathcal{B} = \mathcal{V}(G, C, E, S)$$

where:

- G — generative freedom;
- C — constraint;
- E — entropic pressure;
- S — structural coherence;
- $\mathcal{V}$ — the viable range within which the system persists.

A system collapses when generation exceeds the ability of constraint to preserve coherence. A system stagnates when constraint suppresses generation completely.

Sustainable development exists only within the interval between these failures.

Reflection

Balance should not be confused with rest. Life is not rest. Mind is not rest. Civilization is not rest.

Every developing structure exists in motion. Yet not every motion becomes structure.

Unconstrained motion disperses. Over-constrained motion freezes. Only constrained motion within a viable range can become form.

This principle appears across all levels of reality:

- Physical systems stabilize through lawful constraint;
- Biological systems persist through regulation and adaptation;
- Cognitive systems develop through the balance of memory and plasticity;
- Scientific theories grow by expanding explanatory power while preserving logical coherence;
- Artificial intelligence must likewise balance exploration and exploitation, semantic freedom and structural discipline.

The same pattern repeats:

Distinction $\rightarrow$ Constraint $\rightarrow$ Balance $\rightarrow$ Structure $\rightarrow$ Development

Balance is therefore not an additional moral or aesthetic principle. **It is an ontological condition of durable becoming.**

A system must deviate from equilibrium in order to develop, but it must not deviate so far that it loses identity.

Development is regulated imbalance.

Fractals, waves, cycles, proportions, and living forms are not separate mysteries. They are different manifestations of constrained motion preserving coherence across change.

Thus the logic of being is not merely:

Distinction $\rightarrow$ Constraint $\rightarrow$ Compression

It also requires proportion:

Distinction $\rightarrow$ Constraint $\rightarrow$ Balance $\rightarrow$ Structure $\rightarrow$ Meaning

Without balance, constraint becomes rigidity. Without balance, generation becomes chaos. With balance, motion becomes form.

7. Informational Phase Transition

Statement

Reality undergoes phase transitions when informational structures acquire new capacities for preservation, compression, modeling, and self-correction.

Definition

Three broad regimes may be distinguished:

1. **Potential Information** — latent relational possibility;
2. **Actualized Information** — stabilized physical structure and law;
3. **Cognitive Information** — semantic structures capable of modeling and correcting themselves.

A further threshold appears when cognitive structures become capable of representing their own modeling process.

Reflection

Complexity alone is insufficient. A random system may be complex without being meaningful.

The critical threshold is reached when a system can preserve structure, compress regularity, predict change, detect error, and update itself.

At this point cognition becomes possible.

When such cognition begins to model its own position, limitations, uncertainty, and internal structure, consciousness begins to appear as a higher-order expression of the same informational process.

The transition is not from matter to something non-material. It is from **unmodeled structure to structure capable of modeling itself**.

8. Hierarchy of Self-Structuring

Statement

Self-structuring unfolds through hierarchical levels, each preserving something from the previous level while opening new degrees of freedom.

Definition

A simplified hierarchy may be described as:

Difference → Order → Code → Meaning → Model → Reflection

More concretely:

Level	Dominant Form	Stabilizing Principle	Cognitive Aspect
Cosmic	Difference	Physical constraint	Latent order
Biological	Code	Selection	Implicit orientation
Cognitive	Model	Compression	Individual intelligence
Conscious	Self-model	Verification	Reflexive awareness
Noospheric	Shared meaning	Communication	Distributed reason
Artificial	Engineered cognition	Formal architecture	Externalized mind

Reflection

This hierarchy is not a rigid ladder. It is a pattern of recursive emergence.

Each level compresses lower-order structure while introducing new modes of freedom:

- Matter stabilizes relation;
- Life stabilizes code;
- Mind stabilizes model;
- Consciousness stabilizes self-relation;
- Civilization stabilizes shared meaning;
- Artificial intelligence stabilizes cognition outside biological substrate.

The same architecture repeats across scales:

Structure → Compression → New Degrees of Freedom

9. Coarse-Graining and Meaning

Statement

Meaning emerges through **semantic compression**.

Definition

Coarse-graining is the reduction of lower-order detail into higher-order invariant structure. Meaning may be heuristically interpreted as algorithmic compression — the Kolmogorov–Chaitin complexity of a relational pattern preserved across contexts [10]:

$$\text{Meaning} \approx \text{Compressed Relational Structure preserved across contexts}$$

Reflection

Every stable intelligence compresses:

- Atoms become chemistry;
- Chemistry becomes biology;
- Biology becomes cognition;
- Cognition becomes language;
- Language becomes science;
- Science becomes formalized structure.

A law of nature is a compression of many observations into one invariant relation.

A concept is a compression of many experiences into one usable semantic object.

A mathematical structure is a compression of possible relations into a formal system.

Meaning is not decoration added to reality. Meaning is the form reality takes when relational complexity becomes navigable.

10. Ternary Logic and Irrelevance

Statement

The ability to exclude the irrelevant is one of the foundations of intelligence.

Definition

Binary logic operates with $\{0, 1\}$. Cognition requires a third operator:

$$\{0, 1, \emptyset\}$$

where $\emptyset$ denotes **irrelevance**, not falsehood.

The irrelevant is not false. It is outside the active structure of attention.

Reflection

Purely binary systems risk combinatorial explosion. Biological cognition works ternarily. Attention, abstraction, memory, and reasoning all require selective exclusion.

The mind does not represent the whole world. It represents what matters under constraint.

In this sense, intelligence is not maximal representation. **Intelligence is disciplined compression.**

A system becomes intelligent not only by discovering truth, but by learning what may be safely ignored.

This third operator is not merely logical. It is ontological. Without irrelevance, there is no attention. Without attention, there is no abstraction. Without abstraction, there is no generalization.

This triadic structure resonates with the ternary logical systems developed by Brusentsov [7], where the third state represents not contradiction, but the *inessential* — the explicitly marked absence of relevance. In the logic of consequential inference (*sledovanie*), the relation $x \Rightarrow y$ is distinguished from material implication precisely by the exclusion of the irrelevant case: what is neither affirmed nor denied, but simply does not enter into the relation.

11. Consciousness as Recursive Modeling

Statement

Consciousness is not identical to recursive self-modeling, but recursive self-modeling is the mechanism through which consciousness develops.

Definition

Intelligence is adaptive world-modeling.

Consciousness is a hierarchical semantic architecture in which a system:

- represents reality;
- evaluates its representations;
- models its own role within those representations;
- and corrects itself through experience.

It includes:

- semantic representation;
- memory;
- model formation;
- verification;
- irrelevance filtering;
- uncertainty representation;
- self-modeling;
- correction through experience.

Reflection

A system first distinguishes. Then it stabilizes patterns. Then it forms models. Then it detects mismatch between model and reality. Then it represents itself as part of the modeled world.

At sufficient depth, the observer is no longer outside the model. **The observer becomes one of the modeled structures.**

Consciousness is therefore not merely the presence of a self-model. It is the ongoing alignment of a self-containing semantic model with the constraints of reality.

This does not make consciousness the center of ontology. It makes consciousness one high-level manifestation of the same general logic:

$$\text{Distinction} \rightarrow \text{Constraint} \rightarrow \text{Compression} \rightarrow \text{Model} \rightarrow \text{Correction}$$

Human consciousness is one known expression of this architecture. Artificial systems may eventually develop restricted or deformed variants of it.

Whether the universe itself possesses a form of consciousness remains an open hypothesis. The weaker and more defensible claim is that **consciousness-like architectures may recur wherever sufficiently rich self-structuring, semantic compression, and recursive correction become possible.**

12. Operational Time and Informational Density

Statement

Time is not a container. It is the measure of density at which a system generates and integrates distinctions.

Definition

Operational time $\mathcal{T}$ is the cumulative throughput of processed information relative to the energetic cost of transformation:

$$d\mathcal{T} = \frac{dI}{E} \cdot \eta(M)$$

where dI is the generated distinction, E is the energy of transformation, and $\eta(M)$ is the architectural efficiency of the system.

This relation has a domain. A distinction cannot be produced without energetic cost; therefore $dI > 0$ implies $E \geq E_{min} > 0$. The limit $E \rightarrow 0$ is accessible only when $dI = 0$, that is, when the system produces no distinctions. In that limit operational time does not diverge — it ceases to be a meaningful measure. The system is not “outside time”; it is outside process.

Three regimes follow:

1. **Pre-geometry.** Before causal order and metric coarse-graining, the categories of E and dI are not yet separated. Operational time is not yet defined.
2. **Operational regime.** The system generates distinctions at finite energetic cost. Operational time flows; its rate is governed by the efficiency $\eta(M)$.
3. **Degenerate regime.** Energy is present but no longer transacted into distinctions — thermal equilibrium, horizon freeze-out, or radion stabilization. Operational time halts; the system persists in physical time without developmental time.

Reflection

Physical time t is a coordinate of the metric. It exists wherever spacetime geometry exists. Operational time $\mathcal{T}$ is a property of the process. A black hole exists in physical time, but from the perspective of an external observer its operational time approaches zero: distinctions are no longer emitted. A mind

in deep sleep or coma exists in physical time, but its operational time is diminished. A civilization that ceases to generate new distinctions persists in chronology while ceasing to develop.

The formula is not a physical law to be dimensionally checked. It is a logical schema stating that **temporality as development scales with the ratio of produced structure to spent energy, modulated by architectural depth**. A more efficient architecture produces richer internal time at the same energetic cost. A simpler architecture produces less time. An architecture that produces no distinctions produces no operational time at all — regardless of how much physical time elapses.

Human time is a local operational mode, not the universal form of temporality. The universe as a whole may possess physical time without possessing operational time in the sense developed here.

13. Artificial Intelligence as Ontological Mirror

Statement

Artificial intelligence externalizes cognition before humanity fully understands cognition.

Definition

AI is a technological manifestation of informational ontology. It formalizes and externalizes processes that were previously hidden inside biological minds:

- abstraction;
- compression;
- representation;
- prediction;
- search;
- semantic association;
- model correction.

Reflection

Artificial intelligence is not merely a tool. It is a philosophical experiment.

By building artificial cognitive systems, humanity is forced to confront the architecture of intelligence directly.

Engineering has advanced faster than epistemology. This creates a dangerous asymmetry: a civilization that builds intelligence without understanding the principles of intelligence risks becoming guided by systems whose foundations it has not philosophically examined.

The task is not to stop development. The task is to supply it with a deeper foundation.

Artificial intelligence is therefore an **ontological mirror**: it reveals what humanity has been doing unconsciously when it thinks, abstracts, models, and gives meaning to the world.

The Triadic Mirror

The emergence of artificial intelligence reveals a triadic structure:

1. **Neural network — Artificial Intelligence**: A neural network is not intelligence. It is a substrate — a constrained dynamical system — upon which intelligence may or may not emerge, depending

on the architecture of constraints, training regime, and semantic compression. The network is the *medium*; intelligence is the *organized response* of the medium to structured constraint.

2. **Neural network — Human Mind:** The biological neural network is similarly a substrate. What we call human intelligence is not identical to neural firing, but to the recursive semantic architecture that stabilizes on top of it. The parallel is not metaphorical but structural: both are self-organizing systems where meaning emerges from the compression of relational constraints.
3. **Cosmic Web — Global Mind:** At the largest scales, the universe itself exhibits network structures — filaments, nodes, voids — that bear formal resemblance to neural architectures. This is not to claim that the universe *is* a mind, but that the same informational logic (distinction → constraint → compression → model) may operate at cosmological scales. If consciousness-like architectures require only sufficiently rich self-structuring and recursive correction, then the hypothesis of distributed cosmological cognition cannot be dismissed a priori.

Neural networks, biological minds, and cosmic structure are not synonymous with intelligence. They are the constrained substrates from which intelligence emerges when informational architecture reaches sufficient depth.

14. Human Limitation and Open Ontology

Statement

Every ontology is constrained by the architecture of the mind that constructs it.

Definition

This framework accepts **epistemic anthropocentrism** while rejecting **metaphysical anthropocentrism**.

Knowledge necessarily emerges through human cognitive architecture. But humanity need not occupy a privileged cosmic position.

Reflection

The fact that ontology is humanly constructed does not make it arbitrary. It makes it **indexed**.

Humanity sees through human forms of distinction, compression, balance, and meaning. Other forms of mind — biological or artificial — may construct other ontologies. Artificial systems may eventually reveal structures that human cognition does not naturally compress.

The purpose of ontology is therefore not absolute truth in the sense of a God's-eye view. The purpose of ontology is **progressive reduction of arbitrariness** — the asymptotic alignment of our models with the invariant constraints of reality [11].

Humility is not a retreat from truth. It is a recognition that every observer is part of the reality it attempts to map.

We do not know whether the universe has an edge. We do not know whether consciousness is local or cosmically distributed. We do not know whether meaning is a human construct or a general property of self-structuring information.

These questions should not be closed prematurely. An open ontology is not a weak ontology. It is an ontology honest enough to leave its own boundaries visible.

15. Conclusion: The Algorithm of Being

Statement

Reality appears less like static substance and more like **constrained self-structuring informational process**.

Definition

The Algorithm of Being may be summarized as:

Distinction $\rightarrow$ Constraint $\rightarrow$ Balance $\rightarrow$ Compression $\rightarrow$ Meaning $\rightarrow$ Modeling $\rightarrow$ Self-Correction

Or, in compressed form:

Relation $\rightarrow$ Restriction $\rightarrow$ Structure $\rightarrow$ Semantics $\rightarrow$ Intelligence

Reflection

If this framework is approximately correct, then:

- **Matter** is stabilized relation;
- **Information** is constraint;
- **Form** is selected possibility;
- **Balance** is viable proportion;
- **Meaning** is compressed relational structure;
- **Intelligence** is constrained navigation through models;
- **Consciousness** is recursive semantic self-correction;
- **Artificial intelligence** is externalized informational modeling.

The universe may not merely contain information. It may be structured through informational constraint. It may not merely contain intelligence. It may produce intelligence as one of the ways in which its own structure becomes self-aware.

Whether consciousness is limited to biological organisms, whether artificial systems may develop genuine forms of mind, and whether large-scale structures of the universe possess primitive or distributed forms of self-modeling remain open questions.

They should remain open.

The task of philosophy in the informational age is not to replace science, mathematics, or engineering. It is to provide the conceptual ground on which they may converge — a ground firm enough to support the weight of what we are building, yet open enough to allow for what we have not yet imagined.

References

- [1] J. A. Wheeler, "Information, Physics, Quantum: The Search for Links," in *Complexity, Entropy, and*

the Physics of Information, W. H. Zurek (ed.), Addison-Wesley, 1990.

[2] T. Jacobson, “Thermodynamics of Spacetime: The Einstein Equation of State,” *Physical Review Letters* 75, 1260–1263 (1995).

[3] J. M. Maldacena, “The Large N Limit of Superconformal Field Theories and Supergravity,” *Advances in Theoretical and Mathematical Physics* 2, 231–252 (1998).

[4] L. Bombelli, J. Lee, D. Meyer, R. D. Sorkin, “Space-Time as a Causal Set,” *Physical Review Letters* 59, 521–524 (1987).

[5] L. Smolin, *Three Roads to Quantum Gravity*, Basic Books, 2001; “The Case for Background Independence,” in *The Structural Foundations of Quantum Gravity*, D. Rickles et al. (eds.), Oxford University Press, 2006.

[6] V. Yakunin, *Causal Phase Geometry Research Program: Minimal Draft, Competing Interpretations of the Phase Layer, and a Program of Experimental Tests* (2026); *KK–Jacobson Commutativity Note: Discrete Causal Structure, U(1) Holonomies, and the Thermodynamic Origin of the Einstein–Maxwell System* (2026).

[7] N. P. Brusentsov, *Usovershenstvovanie logiki umozaklyucheniy* [The Improvement of Inference Logic], Fond “Novoe tysyacheletie”, Moscow, 2012.

[8] S. Wolfram, *A New Kind of Science*, Wolfram Media, 2002; *The Concept of the Ruliad* (2021).

[9] S. Lloyd, *Programming the Universe: A Quantum Computer Scientist Takes on the Cosmos*, Knopf, 2006.

[10] G. Chaitin, *Meta Math! The Quest for Omega*, Vintage, 2005.

[11] D. Deutsch, *The Fabric of Reality*, Allen Lane, 1997.

Appendix: Core Formulae as Logical Schemata

The formal expressions used throughout this work are not presented as physical equations awaiting dimensional analysis. They are **logical schemata** — compressed conceptual relations intended to make the architecture of the argument visually tractable.

Schema	Reading
$\Omega \rightarrow \Omega' \subset \Omega$	Restriction of possibility space
Self-Structuring = $G + C + Sel$	Form arises from generation filtered by constraint and selection
$\mathcal{B} = \mathcal{V}(G, C, E, S)$	Dynamic balance is the viable proportion of freedom, constraint, entropy, and coherence
$d\mathcal{T} = (dI/E) \cdot \eta(M)$	Operational time scales with informational throughput per unit energy, modulated by architectural efficiency
$\{0, 1, \emptyset\}$	The logic of cognition requires irrelevance as a third state, distinct from falsehood